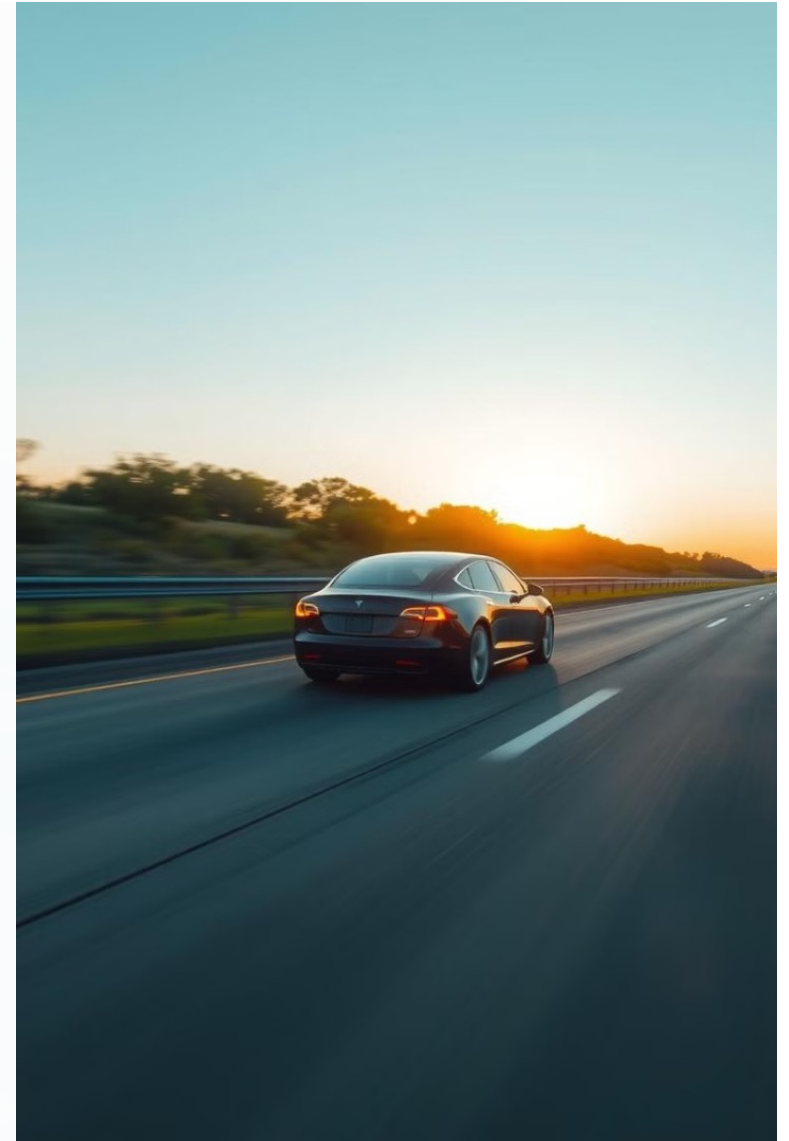
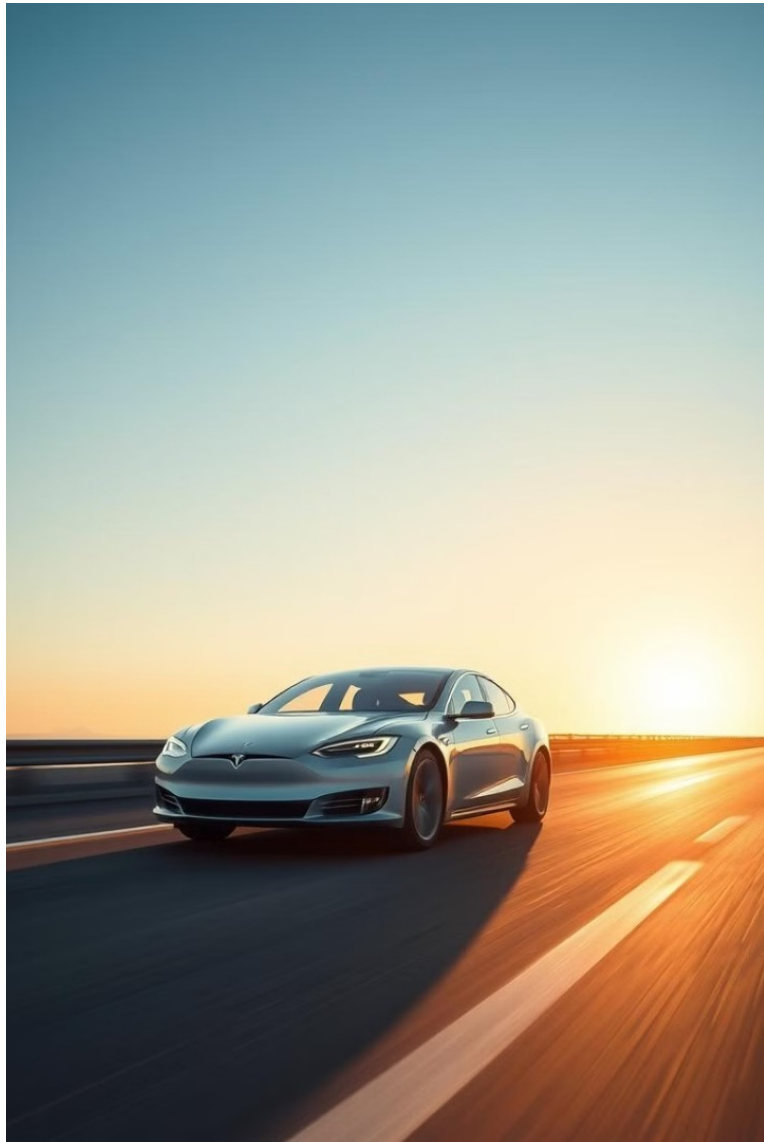


A Study on Tesla Autopilot

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presented by: Jathin N, 1BG21AI043





Introduction

1 Background

Tesla Autopilot, introduced in 2014 with the Model S, marked a significant step towards semi-autonomous driving.

2 Core Features

Autopilot's core features include lane centering, adaptive cruise control, self-parking, and the summon feature.

3 Future Aspirations

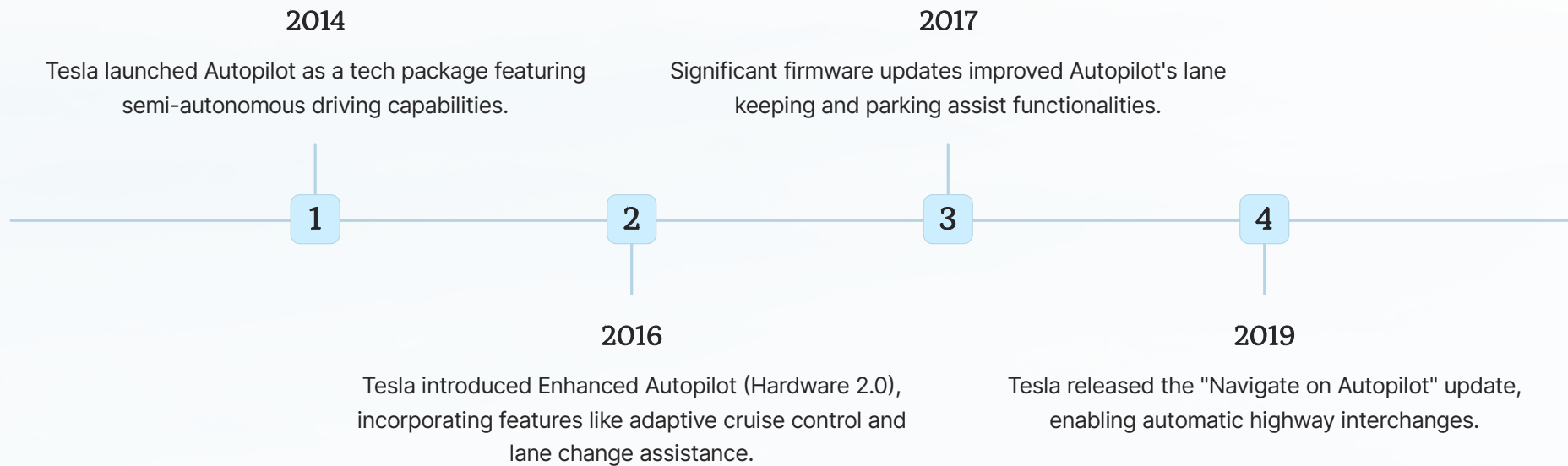
Tesla aims to achieve Full Self-Driving (FSD) with future advancements.

4 Partnerships

Tesla initially collaborated with Mobileye, a leading provider of driver assistance systems, but later transitioned to its own in-house development.



Autopilot Evolution



Key Technologies

Sensors

Autopilot relies on forward-facing radar and cameras for object detection, along with sonars for close-range detection.

Hardware Generations

Autopilot has progressed through multiple hardware generations, starting with Mobileye's EyeQ3 and moving to NVIDIA Drive PX2 and Tesla's custom processors.

Software

Autopilot's software receives continuous wireless updates, incorporating improvements and new features.



Driving Features

Adaptive Cruise Control

Autopilot's adaptive cruise control automatically adjusts speed based on the surrounding traffic.

Lane Departure Warnings

The system alerts the driver if the vehicle veers out of its lane.

Automatic Highway Interchanges

Autopilot can navigate highway interchanges autonomously.

Summon and AutoPark

Autopilot enables the driver to remotely summon the vehicle and automatically park it in tight spaces.



Challenges and Limitations

Regulatory Hurdles

Legal approvals for full autonomy remain a significant challenge.

Driver Responsibility

Tesla emphasizes that drivers must remain engaged and ready to take control at any moment.

Accidents

High-profile accidents involving Tesla vehicles have raised concerns about the safety of Autopilot.

Technology Gaps

Achieving full autonomy requires substantial advancements in various technologies.



Future Outlook

1

Enhanced Autonomy

Future development will focus on expanding Autopilot's capabilities to include city street navigation, traffic light recognition, and stop sign detection.

2

Hardware Upgrades

Tesla will continue to invest in hardware upgrades, such as more powerful processors, to support the increasing demands of autonomous driving.

3

Vision

Tesla envisions a future where autonomous vehicles seamlessly integrate into everyday life, revolutionizing mobility and transportation.

Conclusion



Significant Progress

Tesla Autopilot represents a significant step forward in the advancement of autonomous driving.



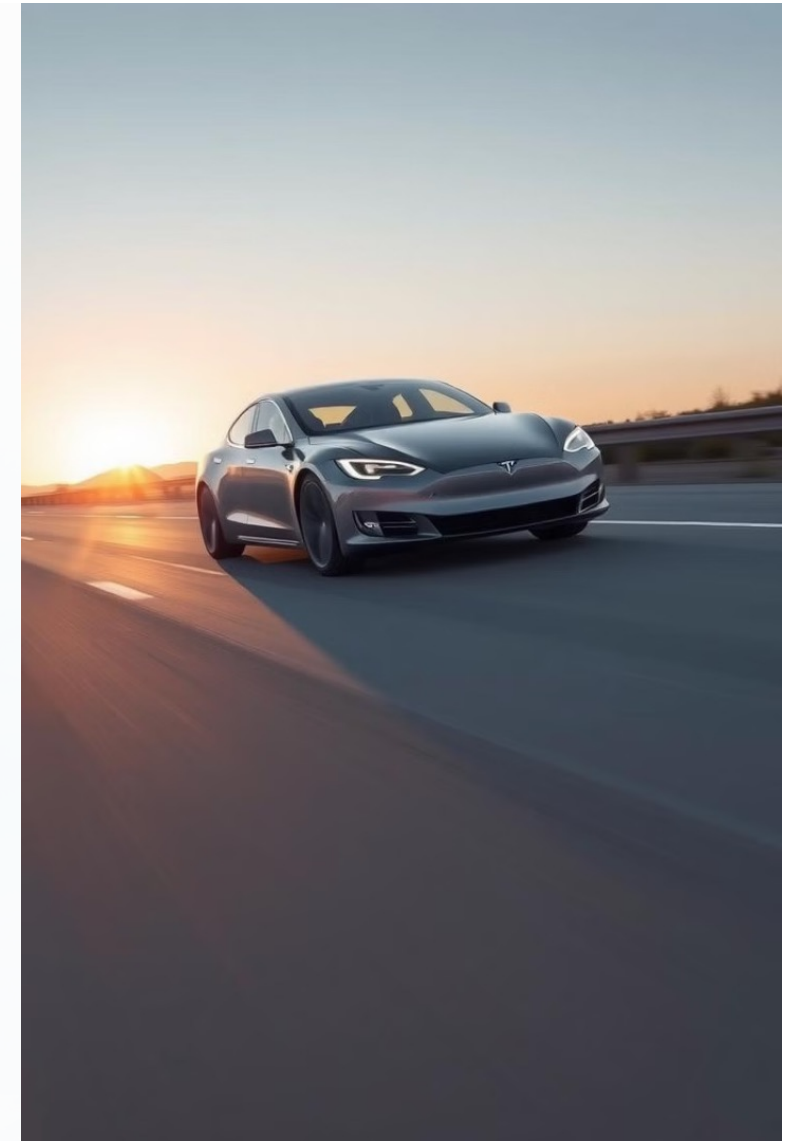
Driver Education

Challenges in driver education and awareness persist, ensuring drivers understand the limitations and responsibilities associated with Autopilot.



Potential for Revolution

With continued advancements, Tesla Autopilot has the potential to revolutionize mobility and reshape transportation.





References

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